

Program 10 : Demonstrate Kubernetes Dashboard and CLI for cluster monitoring

Project Structure

```
├── deployment.yaml
└── service.yaml
```

Step 1 : Create a Deployment.yml file

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab10$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: frontend
          image: nginx
          ports:
            - containerPort: 801rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab10$
```

Step 2 : Create service.yml file

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab10$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab10$
```

Step 3 : Apply the yaml files

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab10$ k apply -f deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab10$ k apply -f service.yaml
service/nginx-service created
```

Step 4 : Verify Resources

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab10$ k logs nginx-deployment-78bb956bd7-7wm4n
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/25 15:36:25 [notice] 1#1: using the "epoll" event method
2025/12/25 15:36:25 [notice] 1#1: nginx/1.29.4
2025/12/25 15:36:25 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/25 15:36:25 [notice] 1#1: OS: Linux 6.14.0-37-generic
2025/12/25 15:36:25 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/25 15:36:25 [notice] 1#1: start worker processes
2025/12/25 15:36:25 [notice] 1#1: start worker process 29
2025/12/25 15:36:25 [notice] 1#1: start worker process 30
2025/12/25 15:36:25 [notice] 1#1: start worker process 31
2025/12/25 15:36:25 [notice] 1#1: start worker process 32
```

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab10$ k describe pod nginx-deployment-78bb956bd7-7wm4n
Name:          nginx-deployment-78bb956bd7-7wm4n
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Thu, 25 Dec 2025 21:06:19 +0530
Labels:        app=nginx
               pod-template-hash=78bb956bd7
Annotations:   <none>
Status:        Running
IP:            10.244.0.23
IPs:
  IP:          10.244.0.23
Controlled By: ReplicaSet/nginx-deployment-78bb956bd7
Containers:
  frontend:
    Container ID:  docker://42d0cce1c0c55769acf42ac6d6fd949d2ec0e2407c7745152d1279761131f377
    Image:         nginx
    Image ID:      docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port:         80/TCP
    Host Port:    0/TCP
    State:        Running
      Started:    Thu, 25 Dec 2025 21:06:25 +0530
      Ready:      True
      Restart Count: 0
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-fkvpr (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers         True
  Initialized                        True
  Ready                             True
  ContainersReady                   True
  PodScheduled                      True
Volumes:
  kube-api-access-fkvpr:
    Type:          Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:    kube-root-ca.crt
    ConfigMapOptional: <nil>
    DownwardAPI:      true
QoS Class:           BestEffort
Node-Selectors:      <none>
Tolerations:         node.kubernetes.io/not-ready:NoExecute on=Exists for 300s
```

Step 5 : Access the services

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:30118

🌐 Opening service default/nginx-service in default browser...

Step 6 : open Kubernetes Dashboard

The screenshot displays the Kubernetes Dashboard interface. The left sidebar contains a navigation menu with categories: Workloads, Service, Config and Storage, Cluster, and Namespaces. The main content area is titled 'Workload Status' and features three large green circles representing the status of Deployments (Running: 1), Pods (Running: 3), and Replica Sets (Running: 1). Below this, there are two tables: 'Deployments' and 'Pods'.

Deployments Table:

Name	Images	Labels	Pods	Created ↑
nginx-deployment	nginx	app: nginx	3 / 3	50 seconds ago

Pods Table:

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created ↑
nginx-deployment-78bb956bd7-5rld	nginx	app: nginx pod-template-hash: 78bb956bd7	minikube	Running	0	-	-	50 seconds ago
nginx-deployment-78bb956bd7-mmdk9	nginx	app: nginx pod-template-hash: 78bb956bd7	minikube	Running	0	-	-	50 seconds ago